



A TMD-oriented Analysis of Charged-Pion Pairs in e^+e^- Collisions

BSc Theoretical Physics Final Year Project

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Abstract

This report investigates whether charged-pion pairs selected from opposite jets in e^+e^- collisions can be used to construct a useful TMD-sensitive observable. The main goal is to test whether the transverse momentum of the selected pion pair, together with the difference between opposite-sign and same-sign yields, can reveal fragmentation structure in a way that is both physically meaningful and comparable to TMD-based theory. Using Monte Carlo event generators, the observable is defined with respect to the thrust axis, and a clear OS/SS separation is found which strengthens as the pion momentum-fraction cut is increased. This behaviour indicates that harder cuts isolate pion pairs that retain a stronger connection to the original fragmentation process and make the underlying fragmentation structure more visible. The resulting distributions are compared with independently generated TMD-based theory predictions, which show the same broad trend and give the most reasonable agreement in the hard-cut, high- z region. While this work is not intended to provide a full extraction of pion TMD fragmentation functions, and such an extraction would in any case require comparison to real e^+e^- data, the additional optimisation studies support the same physical picture and help identify the region in which the comparison is most meaningful. The thesis therefore helps establish this observable as a promising basis for a future extraction using real-world data, and more broadly as a useful step toward more clearly constrained pion TMD fragmentation functions.

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Declaration of Authorship

I declare that all material in this project is my own work, except where there is clear acknowledgement and appropriate reference to the work of others.

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1 Introduction

A central challenge in QCD phenomenology is to identify observables that remain sensitive to the non-perturbative fragmentation process through which partons become hadrons. Ordinary fragmentation functions describe the longitudinal momentum fraction carried by the final-state hadron relative to the fragmenting parton, while TMD fragmentation functions retain additional information on the hadron's transverse momentum relative to the fragmenting parton direction. Because e^+e^- annihilation provides a clean environment with no hadronic initial state, it is one of the most natural places to search for such observables. Previous B-factory measurements showed that hadron yields are sensitive to the distinction between favoured and disfavoured fragmentation^[1]. Here, favoured fragmentation refers to the idea that a parton is more likely to fragment into a hadron whose valence structure contains the same flavour, for example $u \rightarrow \pi^+$, where $\pi^+ = u\bar{d}$, than into one where it does not, such as $u \rightarrow \pi^-$, where $\pi^- = d\bar{u}$. While Belle's measurements of hadron-pair asymmetries showed that unlike-sign and like-sign pion pairs in opposite hemispheres can behave differently in back-to-back events^[2;3]. The possibility of combining these two ideas is what led Valerio to suggest looking into this line of investigation, which then became the basis for the analysis carried out in this thesis.

Motivated by this, the aim of the present thesis is to investigate whether charged-pion pairs selected from opposite jets can define a new observable with meaningful sensitivity to pion TMD fragmentation functions. The analysis focuses on the transverse momentum of the pion pair relative to the thrust axis, and on whether the difference between opposite-sign and same-sign yields becomes more pronounced when the selected pions are required to carry a larger fraction of their parent-jet momentum. This is studied using Monte Carlo simulations of e^+e^- collisions, and the resulting distributions are compared with independently generated TMD-based theory predictions. The thesis then explores several ways of improving that comparison, including scans in the pion momentum-fraction cuts and in z , where z is the longitudinal momentum fraction of the observed hadron with respect to the fragmenting parton, comparisons across perturbative orders, and sensitivity studies of the Monte Carlo fragmentation parameters. While the broader project motivation included the possibility of returning to archived LEP data, the present work is restricted to simulation and theory, and therefore serves as a first step toward a possible future data-driven extraction.

2 Theory

2.1 QCD formulation

2.1.1 QCD as a non-Abelian gauge theory

Quantum chromodynamics (QCD) is the theory of the strong interaction, which describes the behaviour of the partons, (quarks and gluons), contained in hadrons. Unlike quantum electrodynamics (QED), QCD is based on a local SU(3) symmetry.^[4;5] This is due to the fact that quarks carry an extra degree of freedom known as colour, which can take three values, usually labelled red, green, and blue, which is analogous to the primary colours of light. Colour is introduced in QCD because quarks are fermions, so the overall quark wavefunction must obey antisymmetry, and this requires an additional quantum number. The quark fields therefore transform under an SU(3) colour symmetry, with the colour index usually written as $a = 1, 2, 3$. Since quarks and gluons carry colour, the hadrons that are observed must be colour singlet objects, which is why mesons can be written schematically as $q_a \bar{q}^a$, while baryons can be written as $\epsilon^{abc} q_a q_b q_c$, where ϵ^{abc} is the totally antisymmetric tensor.^[4;5] This feature of QCD, that the underlying quark and gluon fields do not appear as free observed particles, is known as confinement.^[6]

The structure of this theory is encoded in the QCD Lagrangian density, as seen in Eq. (1),^[4;5]

$$\mathcal{L} = \bar{\psi}_q^i (i\gamma^\mu) (D_\mu)_{ij} \psi_q^j - m_q \bar{\psi}_q^i \psi_q^i - \frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu}. \quad (1)$$

Local gauge invariance means that the quark fields can be redefined by a local SU(3) transformation at every point in space and time without changing the physics. Since the ordinary derivative ∂_μ does not transform properly under this symmetry, it has to be replaced by the covariant derivative D_μ . This introduces the gluon field into the theory and therefore gives the quark–gluon interaction.^[4] A further difference between QCD and QED appears in the gluon field-strength tensor, which can be seen in Eq. (2),^[4;5]

$$F_{\mu\nu}^a = \underbrace{\partial_\mu A_\nu^a - \partial_\nu A_\mu^a}_{\text{Abelian}} + \underbrace{g_s f^{abc} A_\mu^b A_\nu^c}_{\text{non-Abelian}}. \quad (2)$$

In QCD this tensor contains an extra non-Abelian term which is absent in QED. This means that, in addition to mediating the interaction between quarks, gluons can also interact with one another, unlike photons in QED. The gluon self-interaction can be seen in the Feynman diagram shown in Fig. 1.

It is this gluon self-interaction which leads to many of the important differences between the two theories.

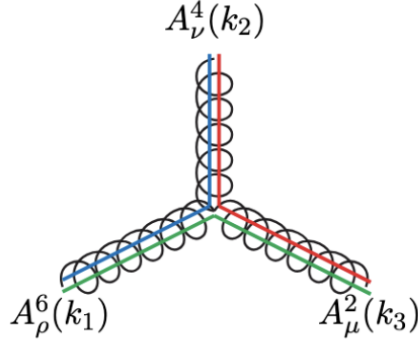


Figure 1: Illustration of a ggg vertex in QCD, where A denotes the gluon field, the superscript gives the colour index, and ρ , ν , and μ are the Lorentz indices.^[5]

In particular, one of the main consequences is that the strong coupling runs with energy scale, which in turn leads to asymptotic freedom at short distances and confinement at large distances.^[4;5]

2.1.2 Running coupling and asymptotic freedom

One may naively define the strong coupling constant in direct analogy with QED, where the interaction strength is set by the fine structure constant $\alpha = e^2/(4\pi)$. In the same way, one defines the strong coupling as $\alpha_s = g^2/(4\pi)$, where g is the QCD coupling constant. However, once QCD is renormalised, it becomes clear that α_s cannot be treated as a constant, and instead depends on the energy scale Q . In the case of e^+e^- annihilation, this hard scale is naturally set by the centre-of-mass energy $\sqrt{s}$.^[7;8]

After renormalisation, the important difference compared to QED is the sign of the running. In QED, the effective coupling increases with the energy scale, and this can be understood intuitively as a screening effect due to virtual e^+e^- pairs in the vacuum around an electric charge. In this sense, the physical QED vacuum behaves like a polarized dielectric medium, as shown in Fig. 2. In QCD, the self-interactions of the gluons spread out the colour charge and generate an antiscreening effect. At one loop, quark loops screen the colour charge, while gluon loops antiscreen it. The gluon contribution dominates, so the QCD beta function is negative overall.^[9] Asymptotic freedom is the antiscreening version of a running coupling. The effective strong charge gets smaller as one probes shorter distances.^[10] This cannot happen in QED, since photons do not carry electric charge, and is a feature of non-Abelian gauge theories.^[11]

Integrating the running of the coupling introduces the scale Λ_{QCD} , so that at leading order the coupling behaves as

$$\alpha_s(Q^2) \sim \frac{1}{\ln(Q^2/\Lambda_{\text{QCD}}^2)}, \quad (3)$$

up to a constant factor. As Q approaches $\Lambda \sim 200$ MeV, the coupling formally becomes infinite, which is known as the Landau pole. The practical interpretation of this is that once Q is not much larger than Λ_{QCD} , α_s becomes large and fixed-order perturbation theory becomes unreliable, so one has to move to a

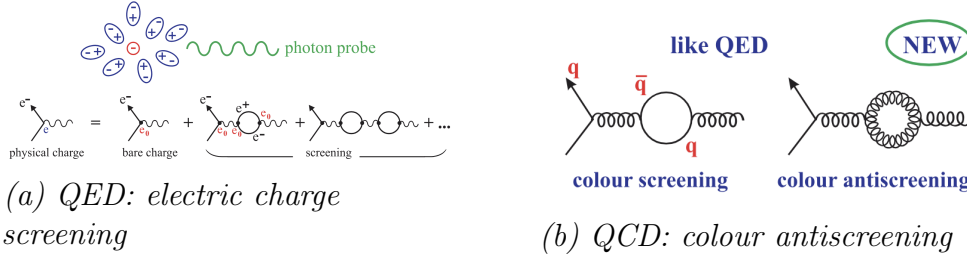


Figure 2: Comparison between electric charge screening in QED and the corresponding QCD picture. The left panel shows vacuum polarization screening an electric charge, while the right panel contrasts a QED-like colour screening contribution with the non-Abelian colour antiscreening effect that arises from gluon self-interactions.^[12]

description involving non-perturbative input. This is broadly what is seen in experiment. Quarks and gluons show up in hard processes, but they are not directly accessible to soft probes.^[10]

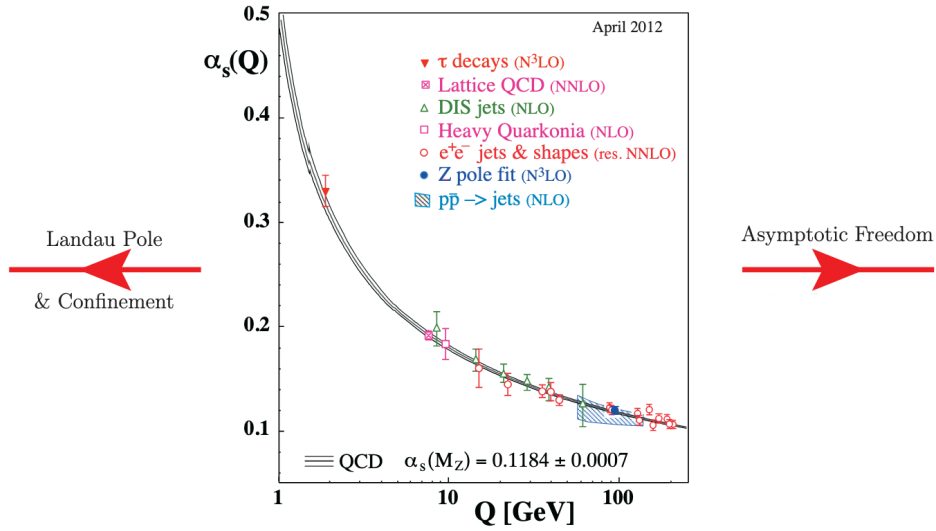


Figure 3: Running of the strong coupling $\alpha_s(Q)$ as a function of the energy scale Q . The graph is read from left to right as moving from low Q , corresponding to long-distance physics, to high Q , corresponding to short-distance physics. At large Q , $\alpha_s(Q)$ becomes small, which is the regime of asymptotic freedom: over very short distances, quarks and gluons inside hadrons behave approximately as weakly interacting partons and can be probed in hard processes. As Q decreases toward the hadronic scale, $\alpha_s(Q)$ grows large, perturbation theory breaks down, and the confinement region is approached, where quarks and gluons are no longer observed as isolated particles.^[5]

One of the main ways this is handled in practice is through factorization, meaning that a cross section σ is separated into a short-distance part, which can be calculated perturbatively, and a long-distance non-perturbative part, which is absorbed into universal objects such as PDFs and FFs, and more recently TMD PDFs and TMD FFs when the transverse momentum of the parton inside the hadron, or of the observed hadron relative to the fragmenting parton, becomes important.^[4;13;14]

2.2 PDFs & FFs

2.2.1 Parton distribution functions and factorization

A hadron's initial state must be described through the use of PDFs (parton distribution functions) $f_i(x, Q^2)$. PDFs give the probability of finding a parton of flavour i (quark or gluon) carrying momentum fraction x at hard scale Q . PDFs however only describe the longitudinal momentum structure of the incoming hadron, but still allow the non perturbative proton structure to enter a perturbative calculation. [15;16]

The cross section σ of a hadron–hadron collision producing a high P_T jet can be written as

$$\frac{d\sigma}{dP_T} \sim \sum_{a,b} \int dx_A f_{a/A}(x_A, \mu_F) \int dx_B f_{b/B}(x_B, \mu_F) \frac{d\hat{\sigma}_{ab}}{dP_T}(x_A, x_B, \mu_F, \mu_R, \alpha_s(\mu_R)). \quad (4)$$

Here, σ denotes the hadronic cross section, while $\hat{\sigma}_{ab}$ denotes the corresponding short-distance partonic cross section. The cross section is the quantity that measures the probability for a given process to occur, while a differential cross section gives that probability as a function of the chosen kinematic variables. [17] This can be done through the use of factorization. In this case, the observed jet begins as a single parton from a short-distance parton–parton scattering, while the probability of finding the incoming partons is encoded in the PDFs. The short-distance physics is contained in the partonic cross section $d\hat{\sigma}_{ab}/dP_T$, while the long-distance structure of the hadrons is absorbed into the PDFs. [15]

The scales μ_F and μ_R enter as the factorization and renormalization scales and can be thought of as setting the resolving power of the probe. If the scale is low, the probe cannot resolve short-distance fluctuations such as a quark splitting into a quark plus gluon in a small spacetime region. As the scale increases, more of the internal structure can be resolved. This scale dependence is described by the DGLAP evolution of the PDFs,

$$\mu_F^2 \frac{d}{d\mu_F^2} f_{a/A}(x, \mu_F) = \int_x^1 \frac{d\xi}{\xi} \sum_b P_{a/b}(x/\xi, \alpha_s(\mu_F)) f_{b/A}(\xi, \mu_F), \quad (5)$$

where $P_{a/b}(x/\xi, \alpha_s(\mu_F))$ is the splitting kernel, which gives the probability for a parton b to split so as to produce a parton a carrying a fraction x/ξ of the parent parton momentum. This shows that the simple valence uud picture of the proton is not enough once the proton is probed at shorter distances. [15]

In practice QCD does not predict the detailed shapes of the PDFs directly, so they are parametrized at a low input scale and then fitted to data, with the evolution to higher Q^2 controlled by DGLAP. Most of the present knowledge of PDFs comes from deep inelastic scattering, especially HERA. [16]

Hadron initial states are therefore much more complicated than e^+e^- . In hadron collisions the proton

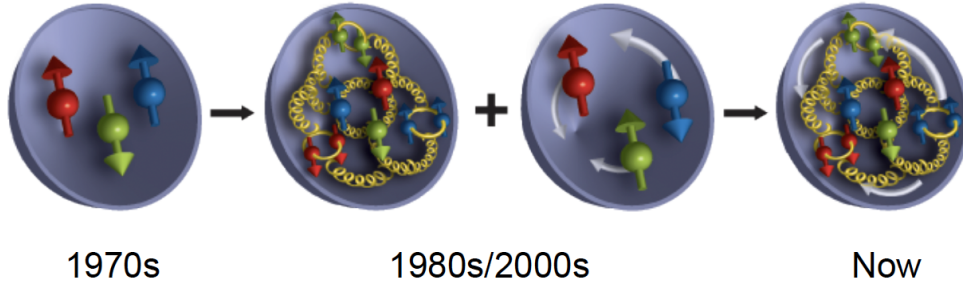


Figure 4: Sketch of how our understanding of hadron structure has evolved, from a simple valence-quark picture to a strongly interacting bound state of quarks and gluons.^[18]

structure has to be modelled through PDFs before the hard process can even be described, whereas in e^+e^- there is no hadronic initial state and no PDFs are needed. This is one of the main reasons why e^+e^- provides a much cleaner environment.^[16]

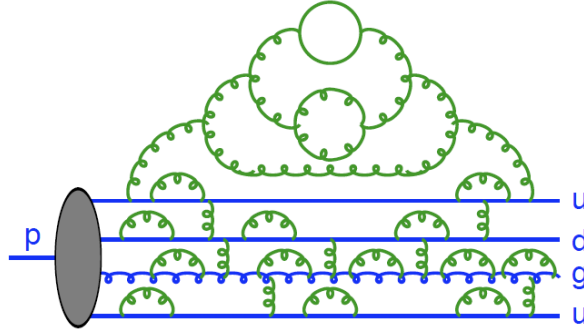


Figure 5: Illustration of partonic fluctuations inside a proton beam, showcasing why hadronic collisions are a more complicated environment.^[5]

Although the partons are predominantly moving parallel to the hadron beam direction, there is no reason to assume that they carry no transverse momentum. In general one expects an intrinsic transverse momentum of order Λ_{QCD} , potentially extending up to a few GeV.^[19]

2.2.2 Fragmentation Functions

Fragmentation functions (FFs) are a crucial building block in perturbative calculations involving observed final-state hadrons, since they describe how the coloured partons produced in a hard process end up as the colourless hadrons that are actually seen in detectors.^[20] They are usually written as $D_i^h(z, \mu^2)$, where D_i^h describes the production of a hadron h from an outgoing parton i carrying a longitudinal momentum fraction z , with μ the resolution or factorization scale. In this sense they are the final-state analogue of the PDFs discussed above. PDFs describe the partonic content of an incoming hadron, while FFs describe the hadronic output of an outgoing parton.^[13;21] This is illustrated in Fig. 6.

As already discussed, e^+e^- annihilation requires no PDFs to describe the initial state.^[13] Because of this, the factorized hadronic cross section can be written in terms of a perturbatively calculable hard part together with non-perturbative fragmentation functions. For single-hadron production this is written as

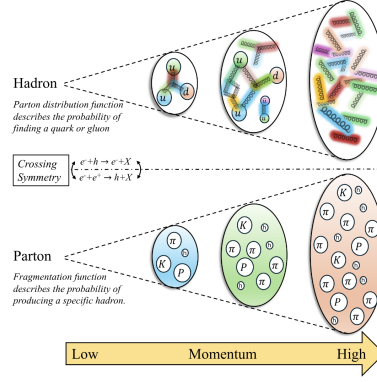


Figure 6: Diagram of the relation between PDFs and FFs. The upper panel shows the partonic content of a hadron, while the lower panel shows the hadronic products generated by a fragmenting parton.^[22]

a convolution of the partonic cross section with the FFs. For two-hadron production the corresponding schematic form is

$$\frac{d\sigma^{H_1 H_2}}{dz_1 dz_2} = \sum_{ij} \sigma_{ij} \otimes D_i^{H_1} \otimes D_j^{H_2} + \sum_i \sigma_i \otimes DD_i^{H_1 H_2}, \quad (6)$$

where the last term contains the double-fragmentation function $DD_i^{H_1 H_2}$, describing the possibility that a single parton fragments into both hadrons H_1 and H_2 . The hadron momentum fractions are defined by

$$z_1 = \frac{2P_1 \cdot Q}{Q^2}, \quad z_2 = \frac{2P_2 \cdot Q}{Q^2}. \quad (7)$$

If the two hadrons are produced in opposite directions, or at least with a clear angular separation, they belong to different jets and can only come from the fragmentation of different partons. In that case the cross section reduces to the product of the ordinary FFs associated with each hadron.^[23]

Certain fragmentation channels are naturally more favoured than others. A favoured fragmentation function describes the fragmentation of a quark into a hadron containing a valence quark of the same flavour, for example $u \rightarrow \pi^+$ and $d \rightarrow \pi^-$, while processes such as $u \rightarrow \pi^-$ are disfavoured.^[3] This means that selecting pion pairs with unlike-sign or like-sign charges gives sensitivity to different combinations of favoured and disfavoured fragmentation functions.^[3] In this way, FFs correlate the observed hadron in the final state with the flavour of the fragmenting parton, and this already suggests that different charge combinations of hadron pairs may behave differently from one another. At the same time, in e^+e^- the flavour structure is not directly accessible event by event, so this sensitivity is indirect rather than exact.^[1]

One important point to note is that the FFs introduced here are still collinear objects and depend only on the longitudinal momentum fraction carried by the observed hadron.

2.3 TMDs

2.3.1 Why TMD fragmentation functions are needed

TMD fragmentation functions are needed once the transverse momentum of the observed hadron becomes important. The ordinary collinear FFs introduced earlier only keep the longitudinal momentum fraction z , while the TMD FFs also keep the transverse momentum generated in the fragmentation process. In this sense they give a more complete description of hadronization, since the hadron is no longer forced to lie exactly along the fragmenting parton direction. In the notation used by Belle, the relevant object is the unpolarized single-hadron fragmentation function $D_1^h(z, k_T, Q)$, where $z = 2E_h/\sqrt{s}$ and k_T is the hadron transverse momentum at the hard scale $Q = \sqrt{s}$.^[24;25]

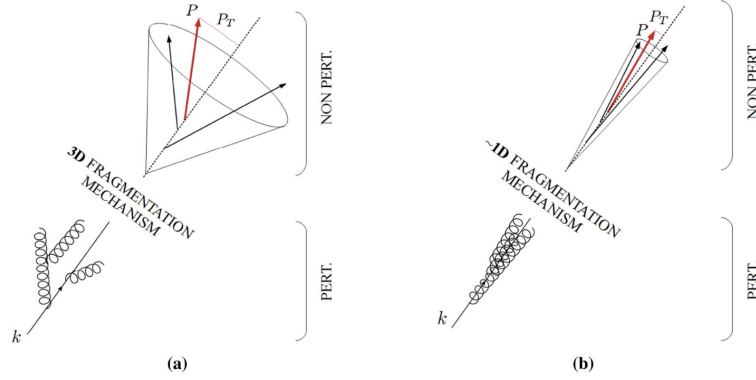


Figure 7: Schematic comparison between the ordinary collinear fragmentation picture and the more general TMD fragmentation picture, where the hadron can carry transverse momentum with respect to the fragmenting parton direction.^[25]

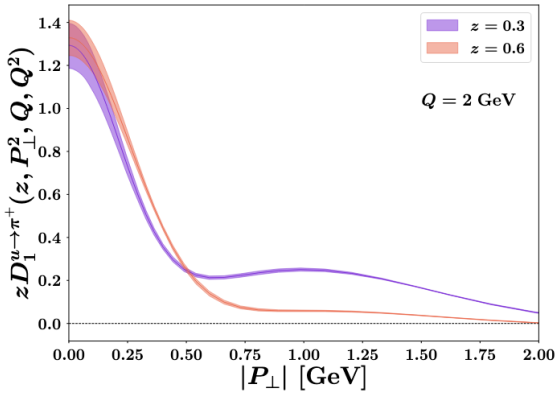
2.3.2 TMD factorization and evolution

The relevant region for this type of physics is the one where there are two scales in the problem: a hard scale Q and a much smaller transverse scale q_T . TMD factorization applies to observables with a hard scale Q and a separate transverse momentum q_T , and in back-to-back hadron production in e^+e^- the near back-to-back region is the one where $q_T \ll Q$. In this regime the cross section is reorganized into a TMD-factorized form involving a perturbative hard part together with two TMD FFs. For dihadron production this can be written schematically as

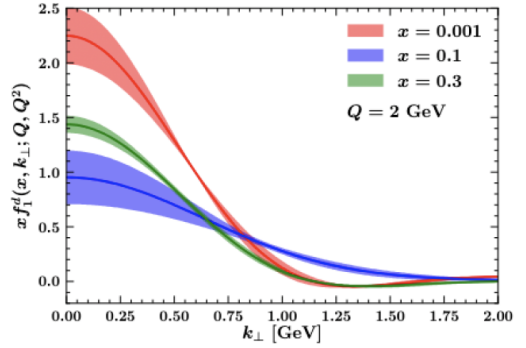
$$\frac{d\sigma^{e^+e^- \rightarrow h_1 h_2 X}}{d^3 P_1 d^3 P_2} \approx \sum_q \hat{\sigma}^{e^+e^- \rightarrow q\bar{q}} \otimes D_{1/e}^{h_1} \otimes D_{2/e}^{h_2},$$

so that the measured distribution is sensitive to the transverse structure of the two fragmentation functions. At the same time, this transverse-momentum dependence is only numerically important in a restricted kinematic region. If an observable is averaged over a broad range of transverse momentum, the detailed

dependence of the TMDs inside the convolution is typically washed out, so that the simpler collinear description, in which the transverse momentum has been integrated over, is usually sufficient. It is only in the low-transverse-momentum region, where $q_T \ll Q$, that the explicit transverse structure becomes numerically significant and TMD factorization becomes the appropriate framework. For this reason, much of the older phenomenology was formulated in terms of ordinary collinear PDFs and FFs rather than TMDs. While recent phenomenological extractions show that the small- q_T region is precisely where TMD sensitivity becomes most useful in practice.^[14;26;27;28] The TMDs themselves are not observed directly. Instead they have to be extracted from measured cross sections, so the phenomenology is an inverse problem in which one deconvolutes the TMDs from the factorized form of the observable.



(a) TMD fragmentation function



(b) TMD parton distribution function

Figure 8: Visualization of a TMD fragmentation function and a TMD parton distribution function. These can be read as momentum-space profiles: low transverse momentum is most probable, the distributions fall at larger transverse momentum, and the different curves show how this profile changes as the longitudinal momentum fraction is changed. This is the visual sense in which TMDs provide a more complete picture of partonic and fragmentation structure.^[28]

In practice it is convenient to work in transverse-coordinate space. The TMD FF can be written as the Fourier transform

$$D_{H/j}(z, z k_T; \mu, \zeta_{FF}) = \frac{1}{(2\pi)^2} \int d^2 b_T e^{-i k_T \cdot b_T} \tilde{D}_{H/j}(z, b_T; \mu, \zeta_{FF}), \quad (8)$$

so that the transverse-momentum dependence is traded for a dependence on the impact-parameter variable b_T . In this form the evolution becomes easier to organize, and the scale dependence is described through TMD evolution and Sudakov-type factors. This is also why the small- q_T region is so important. It is the region where the transverse structure of fragmentation is most visible, but it is also the region where a purely fixed-order treatment is no longer sufficient.^[26;14]

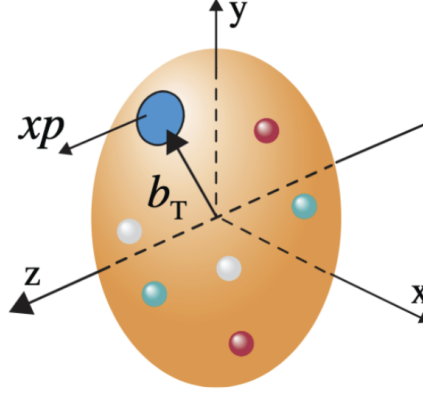


Figure 9: *Illustration of the three-dimensional momentum picture associated with TMDs, where partonic transverse momentum is kept explicitly rather than integrated away.*^[14]

2.3.3 Limitations of available e^+e^- data

Although e^+e^- provides a particularly clean environment for fragmentation studies, there are still practical difficulties in using older data. In many cases the data alone is not enough, and one also needs the software, calibration files, detector documentation, and analysis framework required to read and interpret it correctly. The preservation efforts for JADE, DELPHI, and Belle all show that recovering older e^+e^- data can require dedicated work on archived material, databases, and outdated software environments.^[29;30;31] Even when archived data is available through open-data efforts, making practical use of it is itself a substantial project: one must first access the data, often build an interface to a modern framework such as CERN ROOT or Python, and only then extract and interpret the relevant pion-pair distributions through Monte Carlo comparison. For this reason, a direct LEP-data analysis was beyond the scope of the present thesis, which is instead restricted to simulation and theory as a first step toward a later data-driven study.

2.4 e^+e^- annihilation as the underlying hard process

The process is represented schematically by the Feynman diagram in Fig. 10.

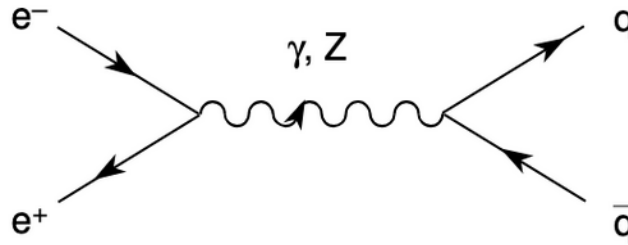


Figure 10: *Feynman diagram for e^+e^- annihilation into two back-to-back jets.*

The electron and positron annihilate into a virtual electroweak boson, represented by γ^* or Z , which then

decays into a quark–antiquark pair,^[32]

$$e^+e^- \rightarrow \gamma^*/Z \rightarrow q\bar{q}. \quad (9)$$

The natural frame for this process is the centre-of-mass frame. In this frame the electron and positron have equal and opposite three-momenta, so the total spatial momentum vanishes and the total available energy is just $\sqrt{s}$. For equal beam energies one has $E_{\text{cm}} = 2E_{\text{beam}}$, so each incoming beam particle carries energy $\sqrt{s}/2$.^[33] At leading order, and neglecting quark masses, the produced quark and antiquark are therefore back-to-back, each carrying energy $\sqrt{s}/2$.^[5] This leading-order picture gives the basic hard process, but it is not yet the full hadronic final state that is observed experimentally. As a result, most leptonic annihilation events are expected to show energy flowing into two back-to-back collimated sets of particles, which are observed as jets.^[6]

2.4.1 Parton showers, hadronization, and jet formation

The leading-order prediction for the process is a simple two-parton state, but this is not what is observed experimentally.^[5] This is due to the fact that quarks and gluons are coloured particles, and so they undergo QCD radiation.^[5] The initial quark–antiquark pair does not remain a simple two-parton state. Instead, it develops into a larger number of partons via branchings such as the emission of a gluon or the production of a quark–antiquark pair via the splitting of a gluon.^[5;32] This is called the parton shower.

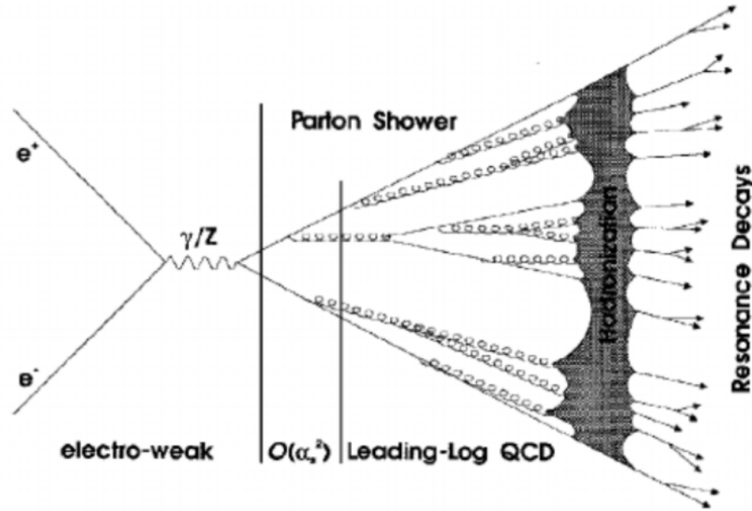


Figure 11: Sketch of the development of an e^+e^- event from the electroweak hard process through the parton shower, hadronization, and resonance decays.^[34]

^[5;7] The hard scattering process therefore gives the initial partons, while the QCD radiation gives a more realistic partonic final state. This continues until the perturbative description is no longer reliable.^[5;7] At that point the hadronization process has to begin, since coloured particles are not observed.^[7] The

perturbative partonic state must then be replaced by a non-perturbative hadronic state. In practice this transition is described by phenomenological models, most notably the Lund string model and the cluster model.^[7;32] After the hadronization stage, the energetic flow produced by the original partons is what is observed. This is what is called a jet. A jet can therefore be considered to be the hadronic remnant of a parton.^[5;7] At higher orders, hard gluon emission can also produce three-jet event topologies, so the simple back-to-back dijet picture is only the leading-order limit.^[32;7]

2.4.2 Jet topology and event structure

The hadronic state can then be characterized by the topology of the jets and the event. Event-shape variables can be used to measure whether the event is pencil-like, planar, or spherical.^[7] One of the most important event-shape variables is the thrust T , defined as

$$T = \max_{\mathbf{n}} \frac{\sum_i |\mathbf{p}_i \cdot \mathbf{n}|}{\sum_i |\mathbf{p}_i|}, \quad (10)$$

where $\mathbf{p}_i$ are the momenta of the observed hadrons.^[7] If the event is dominated by two jets back to back, the thrust will be close to 1. The thrust axis is defined as the direction $\mathbf{n}$ which maximizes the thrust.^[7]

Sphericity S provides a more global measure of how isotropic the event is. In a pencil-like two-jet event the sphericity will be small, while larger values correspond to a more spherical final state.



Figure 12: *Left: Illustration of a dijet event aligned along the thrust axis. Right: Illustration of a more isotropic multijet event, showing the type of event topology that leads to a larger sphericity.*^[35]

The thrust axis is particularly important because it provides a natural axis for the final state. In the literature, hadron transverse momentum measured with respect to the thrust or jet axis has been identified as a possible TMD-sensitive observable in e^+e^- collisions. At the same time, the jet axis is not trivial to reconstruct, and the resulting transverse-momentum distributions are very sensitive to how well that axis is determined, while the identification of the quark axis with the thrust or jet axis also becomes problematic beyond leading order.^[21;36] The momentum transverse to the jet axis is therefore one of the most direct

channels for the phenomenological extraction of TMD fragmentation functions.^[36]

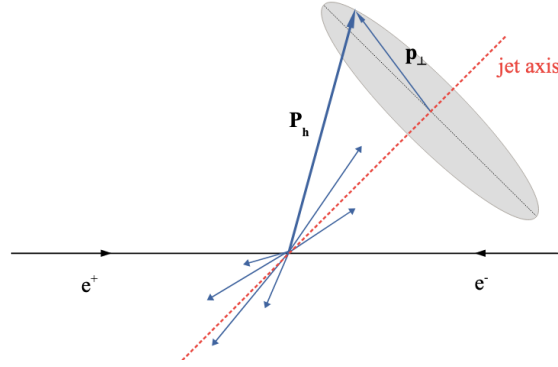


Figure 13: Diagram of hadron transverse momentum measured with respect to a reconstructed jet axis in e^+e^- annihilation, showing the hadron momentum P_h and its transverse component $p_\perp$ relative to the jet axis.^[36]

2.5 Monte Carlo event generators

2.5.1 How Monte Carlo event generators work

The need for the use of Monte Carlo methods arises from the fact that the realistic hadronic final state is not solely determined by the hard process.^[5] After the inclusion of parton showering, hadronization, and decays, the number of particles in the final state becomes large and the phase space becomes enormous.^[5] This is a problem in collider physics, since the dimensionality of the final-state phase space increases rapidly as the number of particles increases.^[5] For example, the convergence rate of the trapezoidal rule and Simpson's rule scale like $1/n^{2/d}$ and $1/n^{4/d}$ respectively, where d is the dimension of the integral.^[5] The problem is avoided by the fact that the Monte Carlo method has the characteristic $1/\sqrt{N}$ convergence, making it much more suitable for these high-dimensional calculations.^[5]

The second reason for the use of Monte Carlo generators is the need for an exclusive final state rather than just an inclusive cross section.^[27] In order to describe the full event, a fixed-order partonic calculation alone is not sufficient.^[27] The hard process, the parton shower, the hadronization stage, and the decays of unstable particles all need to be included.^[5;27] This is achieved by using event generators to produce realistic events which can then be studied at hadron level.^[27] In this way, Monte Carlo event generators give a numerical description of the transition from the basic hard process to the full hadronic final state, while still keeping track of the event structure.^[5]

2.5.2 PYTHIA

PYTHIA is a general-purpose event generator. The hadronization process in PYTHIA is based on the Lund string model. In this picture, the colour field connecting the partons is treated as a string which then

breaks repeatedly into new $q\bar{q}$ pairs as the partons move apart.^[7;32;37] The hadrons are then formed from these repeated string breakings. The transverse momentum generated in this process is sampled from a Gaussian distribution, motivated by the tunnelling picture of pair production in the string field.^[7;38]

Energy density of color fields

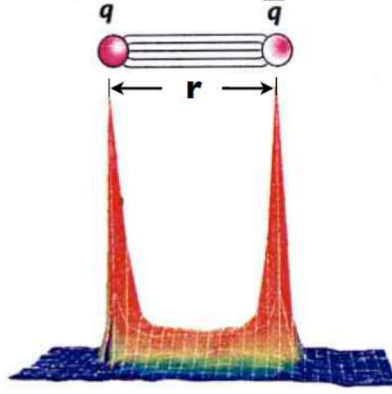


Figure 14: Illustration of the colour field concentrated into a flux tube between a quark and an antiquark. As the separation increases, energy is stored in the string-like field, providing the physical picture behind string breaking and the tunnelling production of new $q\bar{q}$ pairs.^[39]

The hadronization process in PYTHIA is therefore not just the attaching of hadrons to partons, but an iterative process of string breaking with both longitudinal and transverse momentum generation.^[37;38]

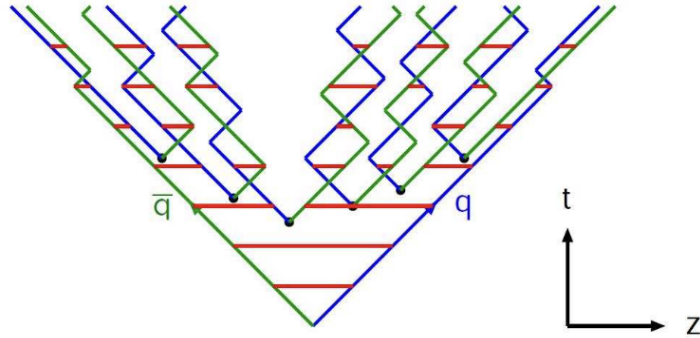


Figure 15: Schematic of the motion and repeated breakup of a $q\bar{q}$ string system in the Lund string model. The blue and green lines show the trajectories of quarks and antiquarks in the t - z plane, while the red segments show the string stretched between neighbouring partons. As the original $q\bar{q}$ pair moves apart, the string breaks repeatedly through the creation of new $q\bar{q}$ pairs, producing the chain of hadrons in the Lund model.^[40]

A useful feature of the PYTHIA program is that the complete history of the event is contained in the event record.^[37;41] Each particle is stored together with information such as the identity code, status code, mother and daughter indices, colour tags, and four-momentum.^[41] This allows the event to be traced step by step through the shower and hadronization history.^[42;41]

PYTHIA also contains a number of tunable parameters which affect the shape of the final hadron distributions.^[38;43] For the present work, one of the most important is `StringPT:sigma`, which controls the width of the transverse momentum generated when the string breaks.^[38] Another relevant parameter is

the effective α_s used in the timelike shower, since this changes the strength of the perturbative radiation before hadronization.^[44] In addition to individual parameters, PYTHIA also uses global tunes, which are fitted parameter sets chosen to give an overall description of the data.^[45;43] The important point is that these parameters do not all affect the same stage of the event: some act on the perturbative shower, while others act on the non-perturbative hadronization process.^[38;44;45;43]

Although Monte Carlo tools incorporating TMD dynamics do exist, such as CASCADE3 and KATIE, they are aimed mainly at hadron-collision studies and initial-state TMDs rather than at the type of e^+e^- analysis considered here. Since a fully general TMD treatment is still not available, they are not yet used as standard general-purpose generators in the same way as PYTHIA or HERWIG. It has nevertheless been argued that PYTHIA effectively includes several elements of TMD physics through its initial- and final-state showering.^[46;47;48]

3 Methodology

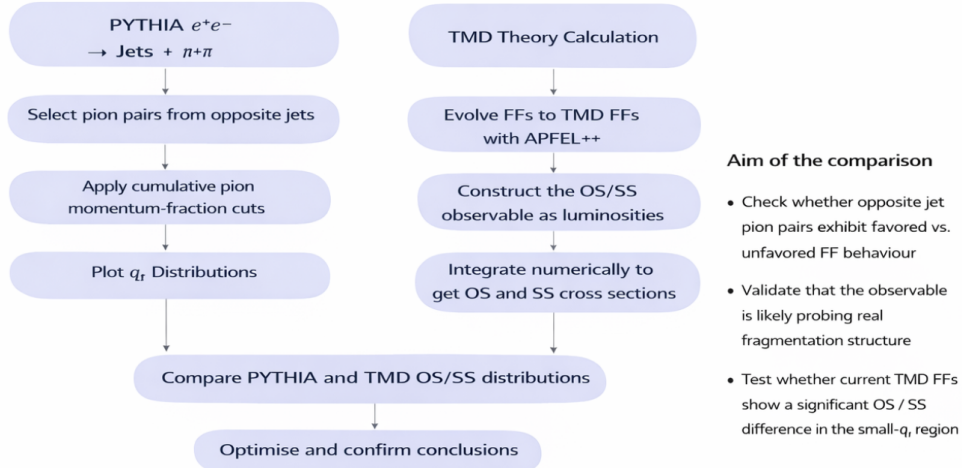


Figure 16: Flowchart summarizing the methodology of this thesis. The PYTHIA event-selection path and the independent TMD-theory calculation are carried out separately before being combined in the OS/SS comparison and later optimisation studies.

3.1 Simulation Setup

An electron beam and a positron beam are made to collide at a centre-of-mass energy of 91.2 GeV, corresponding to the Z -pole. This energy is chosen because it provides a clean high-energy environment in which the dominant hard process is $e^+e^- \rightarrow Z/\gamma^* \rightarrow q\bar{q}$, leading naturally to a dijet final state. These collisions are simulated primarily using the Monte Carlo event generator PYTHIA.

The baseline sample is generated with standard PYTHIA settings. However, some shower and hadronization

parameters are varied later in the optimisation study. The event samples used in this thesis are not all of the same size: the main cut-based analysis used much larger event sets, in some cases up to 100 million events. Where relevant, the number of generated events is stated explicitly for the corresponding plot or study.

The random seed was initially chosen to be 123456788 for reproducibility. In later stages of the analysis, the setup was adjusted where needed to allow multithreading and speed up event generation.

3.2 Thrust Axis and Jet Reconstruction

All the final-state visible particles are used as input to the jet reconstruction program, and the jets are reconstructed using the anti- k_T algorithm in FASTJET^[49] with a radius parameter of size $R = 0.4$. The thrust is also calculated using the same set of final-state particles according to Eq. (10). Here, we scan over a number of trial directions n on a unit sphere using a 50×50 grid in the polar and azimuthal angles and compute the thrust for each direction, then select the direction that maximizes the above quantity. This direction is then used as the thrust axis and as the event axis in the calculation of the pion pair observable. This is important because, in e^+e^- , the dijet system has no preferred orientation relative to the beam, so the event axis should follow the actual jet topology.

The anti- k_T algorithm clusters particles using the distance measures

$$d_{ij} = \min \left(\frac{1}{p_{Ti}^2}, \frac{1}{p_{Tj}^2} \right) \frac{\Delta R_{ij}^2}{R^2}, \quad d_{iB} = \frac{1}{p_{Ti}^2}, \quad (11)$$

where $\Delta R_{ij}^2 = (\eta_i - \eta_j)^2 + (\phi_i - \phi_j)^2$. Here η is the pseudorapidity, a way of describing angle relative to the beam, and ϕ is the azimuthal angle around the beam. Particles with the smallest distance are clustered first until all particles are assigned to jets. Since hard particles dominate this distance measure, the final jets are usually circular and cone-like in the (η, ϕ) plane.

To ensure the event is dijet-like and thus suitable for analysis, several cuts are applied at the event level. The event must satisfy $T > 0.8$, exactly two jets with $p_T > 5$ GeV must be reconstructed, and the two jets must be close to back-to-back with $|\Delta\phi| > 2.8$ rad. These cuts are important because the pion-pair observable is only meaningful if the event really has a clean back-to-back dijet structure. They define the working event sample used in the later pion-pair analysis and in the baseline event-shape checks.

3.3 Definition of pion pair observable

The charged pions within each of the two selected jets are identified. These pions are required to be traceable through the PYTHIA event record back to one of the original quarks from the $q\bar{q}$ pair. Physically, the hadronic final state comes from this original $q\bar{q}$ system, but this requirement is still imposed so that the selected pions can be cleanly associated with that hard process within the event record.

Every possible cross-jet pion pair is then formed by taking one accepted charged pion from the first jet and one accepted charged pion from the second jet. If the momentum of the pion selected from jet 1 is $\vec{p}_{\pi_1}$ and the momentum of the pion selected from jet 2 is $\vec{p}_{\pi_2}$, the pair momentum is defined as the three-vector sum

$$\vec{q} = \vec{p}_{\pi_1} + \vec{p}_{\pi_2}. \quad (12)$$

The transverse momentum of the pair is then calculated with respect to the thrust axis,

$$q_T = |\vec{q} - (\vec{q} \cdot \hat{n}_T)\hat{n}_T|. \quad (13)$$

The selected pair is taken to be the one which maximises this q_T . This choice is intended to favour pion pairs that remain strongly associated with the leading jet structure while still retaining a sizeable transverse component, and therefore provide a more sensitive probe of the non-collinear part of fragmentation.

3.4 Pion Momentum Fraction and Histogram Construction

Once the final charged pion pair has been selected in each event, it is classified according to whether it is opposite-sign or same-sign. An opposite-sign pair is one in which the two selected pions have opposite electric charge, while a same-sign pair is one in which the two selected pions have the same electric charge. Since only one charged pion pair is selected in each event, every accepted event contributes only once to the later analysis and is therefore counted as either OS or SS, but not both.

For each pion in the selected pair, its momentum is then compared to the momentum of the jet from which it was selected. This is done through the ratio $|\vec{p}_\pi|/|\vec{p}_{\text{jet}}|$, which is used as the pion momentum fraction. The histograms are filled cumulatively, so any event that passes a tighter cut is also included in all looser cuts. For example, if both selected pions satisfy the 40% cut, then that event is also included in the uncut and 20% histograms, but not in the 60% histogram. In this way, tighter cuts select pion pairs that carry more of the jet momentum and are therefore more closely tied to the leading structure of the two jets, which is expected to retain the most information from the original quark before it is diluted in the later

stages of fragmentation.

The raw event histograms show the number of accepted OS or SS events in each q_T bin. In the main cut-dependent analysis, these q_T distributions are constructed using 40 bins, unless stated otherwise for a specific earlier confirmation plot.

3.5 TMD Theory Calculation and Comparison

The TMD theory calculation used in this work is based on code provided by Valerio Bertone and Matteo Cerutti, with parts of the implementation adapted for the present analysis. The starting point is a set of ordinary collinear pion fragmentation functions, which are then evolved into TMD-based fragmentation functions using APFEL++.^[50;51;52] As mentioned earlier, the theory comparison is carried out using the variable z , where z is the longitudinal momentum fraction of the observed hadron with respect to the fragmenting parton. This is not exactly the same as the jet-based pion momentum fraction used in the Monte Carlo analysis, since that quantity is defined through the ratio of the pion momentum to the reconstructed jet momentum. However, it plays a similar role in that both are intended to select more leading hadrons. For further information on the code and its construction, the reader is referred to the epemTMD repository: <https://github.com/mcerutti996/epemTMD>.

The motivation for using this comparison is therefore not to obtain a one-to-one fit to the PYTHIA distributions. Instead, the aim is to test whether an independent TMD-based calculation shows the same broad OS/SS picture that appears in the Monte Carlo. If it does, this strengthens the argument that the observable is probing genuine fragmentation structure and is not simply an artifact of PYTHIA. In this sense, the theory comparison is being used as a consistency check on the physical idea behind the observable, rather than as a completed extraction.

4 Results

4.1 Testing the OS/SS sensitivity hypothesis

As already discussed, the main goal here is to test whether charged-pion pairs selected from back-to-back jets show a clear difference depending on whether they are opposite-sign (OS) or same-sign (SS). This expectation comes from the favoured and disfavoured fragmentation picture described earlier. In practice, that distinction is expected to show up most clearly at high z , where the observed pion carries a large fraction of the parent parton momentum and therefore remains more directly connected to the original fragmentation process. If the selected pions are closely connected to the original quark and antiquark,

then OS and SS pairs should not appear with the same yield.

The results of this test are shown in Figure 17. These distributions were obtained from 100 million PYTHIA events, using only events that pass the selection cuts described at the end of Section 3.2. In each accepted event, all allowed charged-pion pairs are considered and the pair that maximises q_T is selected. That selected pair is then classified as either OS or SS, so each successful event contributes only once. The four panels correspond to different cumulative cuts on the pion momentum fraction. The uncut case includes all selected pairs, while the 60% panel contains only pairs in which both selected pions pass the 60% requirement. Since the cuts are cumulative, every event that appears in the 60% sample is also contained in the lower-cut samples.

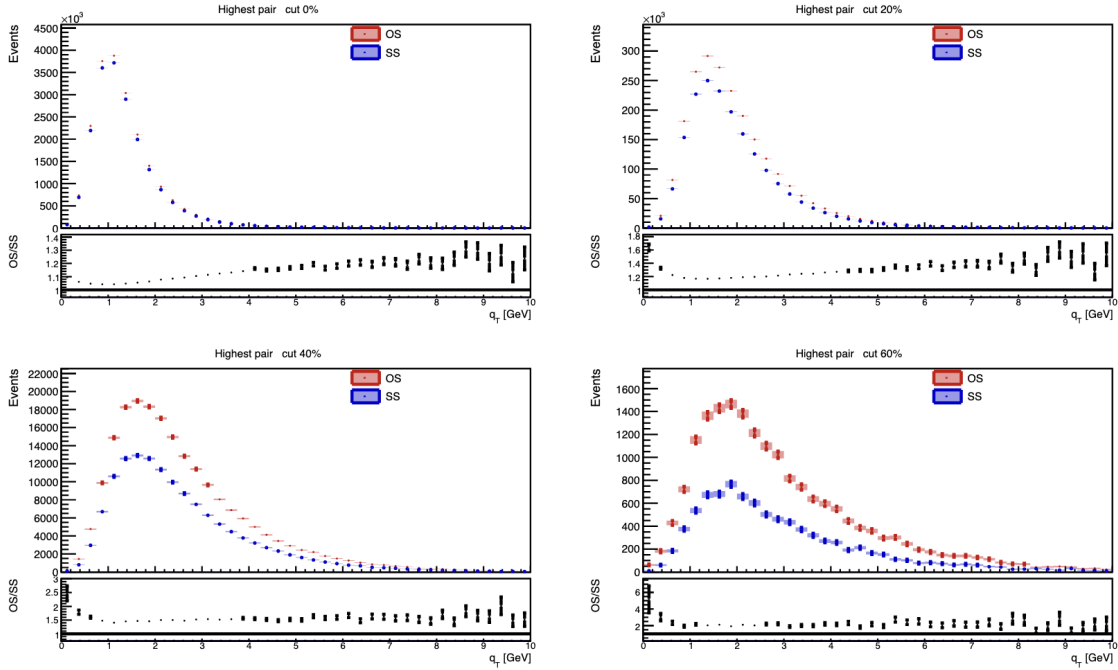


Figure 17: *Cut-dependent OS and SS q_T distributions for the final selected pion pair. In each accepted event, all allowed charged-pion pairs are considered and the pair that maximises q_T is chosen (this is what “highest pair” refers to in the plot titles), so each event contributes only once. The four panels correspond to cumulative cuts on the pion momentum fraction, with the uncut panel including all selected pairs and the higher-cut panels containing subsets of that sample. In the upper plots, the red and blue points show the OS and SS event counts in each q_T bin. In the lower plots, the bin-by-bin OS/SS ratio is shown in order to make the charge-dependent yield difference more visible.*

In the upper part of each panel, the OS and SS q_T distributions are shown as event counts in each q_T bin. The vertical uncertainties come from Poisson counting statistics in each bin, while the horizontal width simply reflects the bin width. In the lower part of each panel, the bin-by-bin OS/SS ratio is shown, with the uncertainty obtained by propagating the Poisson counting uncertainties on N_{OS} and N_{SS} . This lower plot is useful because it makes the relative OS excess easier to see than in the raw histograms alone.

A first general feature of Figure 17 is that most events lie in the small- q_T region, with the distributions

rising to a peak and then falling away toward larger q_T . The more important feature, however, is the changing separation between OS and SS as the pion momentum-fraction cut is tightened. In the uncut case, and still to a large extent at 20%, the OS and SS distributions remain fairly close. By 40%, a clear separation has appeared, with the OS yield lying above the SS yield across much of the distribution. By 60%, this becomes even more pronounced, with the OS sample typically around a factor of two larger than the SS sample over much of the populated region. The main point of Figure 17 is therefore that the OS excess becomes much clearer once harder pion momentum-fraction cuts are applied.

At the same time, the distributions also broaden as the cut is increased. The selected sample is therefore not only becoming more separated in OS and SS yield, but is also moving toward higher q_T . This matters because the tighter cuts are not simply removing events at random. Instead, they are selecting a more specific subset of pion pairs, namely pairs that are more leading inside their jets while still retaining non-negligible transverse momentum. Note that the term leading here refers to a pion that contains a large fraction of the jet momentum and is therefore more likely to retain inherited information from the original quark, such as longitudinal momentum and some of the transverse-momentum structure generated during fragmentation.

This trend is summarized more compactly in Figure 18. The left panel shows the total ratio $N_{\text{OS}}/N_{\text{SS}}$ as a function of the pion momentum-fraction cut, from 0% up to 90%. The ratio rises steadily with increasing cut, showing that the OS excess does not appear only at one isolated value but strengthens systematically as the selection becomes harder. The right panel shows the mean $\langle q_T \rangle$ of the selected OS and SS distributions after each cut is applied. For both charge combinations, the mean q_T increases as the cut is tightened, confirming the broadening already seen in Figure 17. The OS points also remain slightly above the SS points across most of the cut range, indicating that the OS sample has a slightly larger average transverse momentum as well as a larger overall yield.

The physical interpretation of this behaviour is that harder momentum-fraction cuts pick out more leading pions, and these are the pions for which the favoured versus disfavoured fragmentation difference is expected to show up most clearly. At low momentum fraction, the selected pion is less likely to be closely tied to the original hard quark from the original $q\bar{q}$ pair. Instead, it can come from softer later-stage jet structure, where there are many more partons and more hadronization activity contributing to the final state. In that region, the flavour information from the original quark is less cleanly preserved, so the favoured/disfavoured difference gets washed out. By contrast, a more leading pion is more likely to retain a stronger connection to the initiating quark, so the OS/SS difference is expected to appear more clearly. Since the analysis also selects the pair that maximises q_T , the final sample is biased toward pions that are both relatively leading

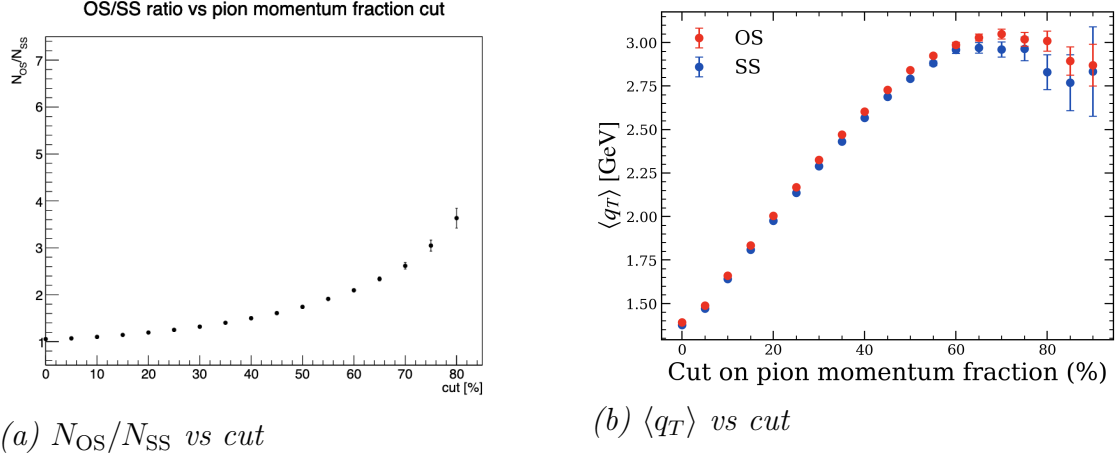


Figure 18: Compact summaries of the cut dependence of the selected pion-pair sample. The left panel shows the total OS/SS yield ratio as a function of the cumulative pion momentum-fraction cut, confirming that the OS excess becomes stronger as the cut is tightened. The right panel shows the mean $\langle q_T \rangle$ of the selected OS and SS samples after each cut is applied, showing that the distributions broaden at the same time. The OS points remain slightly above the SS points across most of the cut range, although the statistical uncertainties grow at the highest cuts because of the reduced event count.

and still non-collinear, which is exactly the region where TMD-sensitive behaviour is most interesting.

There is, however, an important statistical limitation. At the highest cuts the accepted sample becomes much smaller, so the statistical uncertainties increase noticeably. This is already visible in the ratio plots of Figure 17 and in the larger error bars at the highest-cut points in Figure 18. The trend itself remains clear, but its overall precision becomes weaker once the cuts are made too hard. The useful region is therefore not simply the hardest cut possible, but the region where the cut is strong enough to expose the OS/SS structure while still leaving enough statistics for a meaningful comparison.

Overall, these results give strong Monte Carlo support to the original OS/SS sensitivity hypothesis. Once the analysis is restricted to more leading pion pairs, the OS yield becomes systematically larger than the SS yield, and this difference grows with the pion momentum-fraction cut. This is consistent with the idea that the selected observable is sensitive to the underlying fragmentation structure. By itself this does not yet prove the effect in real data, but it does show that the hypothesis has a clear and physically sensible basis in simulation.

4.2 Connecting the observed OS/SS trend to TMD theory

Now that the PYTHIA results have shown a clear separation between OS and SS at the harder cuts, the next step is to check whether the TMD theory gives this same kind of separation. The main reason this check matters is that these theory curves are built from the TMD fragmentation-function setup discussed in Section 3.5, so if a clear OS/SS difference appears here then that difference is coming from the favoured versus unfavoured fragmentation structure rather than from some unrelated feature of the Monte Carlo. In

that sense, this is the first check that the OS/SS separation seen in PYTHIA is actually connected to the fragmentation picture we are trying to probe.

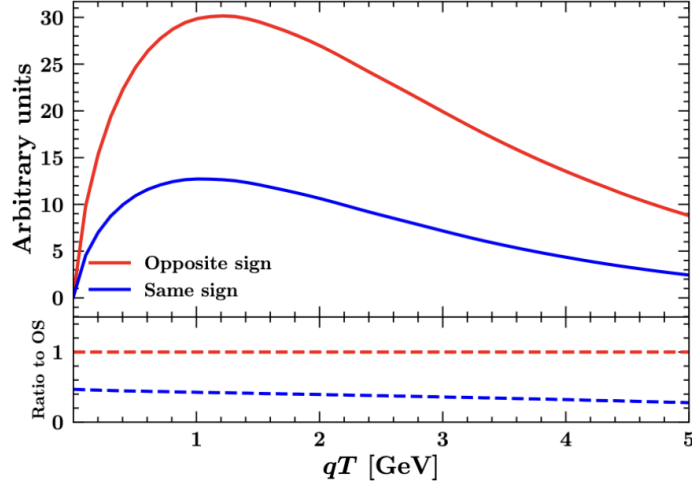


Figure 19: *Standalone TMD-theory prediction for the OS and SS q_T distributions at $z = 0.7$ at NNNLO. The upper panel shows the independently generated OS and SS theory curves in arbitrary units, while the lower panel shows the SS/OS ratio in order to make the charge-dependent separation more visible. The clear difference between the two curves arises within the TMD fragmentation-function setup itself, before any comparison to PYTHIA is made.*

The standalone theory result is shown in Figure 19. This plot shows the OS and SS TMD theory curves as functions of q_T , with the OS curve in red and the SS curve in blue. The lower panel is included simply to make the difference between the two easier to see. Since the comparison is made relative to the OS curve, the red dashed line is fixed at 1, while the blue dashed line shows the SS/OS ratio. The theory curves are evaluated at next to next to next to leading order (NNNLO) and are given in arbitrary units, with the y -axis running up to about 30, and in this standalone version the q_T range is shown up to 5 GeV. This will later be extended to 10 GeV for the overlay comparison with PYTHIA.

The main thing to notice in Figure 19 is that there is already a clear separation between OS and SS in the theory alone, with the OS curve sitting above the SS curve across the plotted range. At the same time, both curves show the same general pattern in q_T , rising in the small- q_T region and then falling away at larger q_T . Since this result is built from the TMD fragmentation-function setup, this separation is exactly the kind of effect expected from the favoured versus unfavoured fragmentation picture. It is also important that this theory result is evaluated at $z = 0.7$. Since the maximum possible value is 1, this is already a relatively high momentum fraction, and that is the region where the favoured versus unfavoured difference is expected to show up most clearly. This makes it the most natural theory setting to compare against the harder-cut PYTHIA samples, where the selected pions are also expected to be more leading in their jets.

This theory result is then overlaid on top of the PYTHIA distributions in Figure 20. Since the theory is in arbitrary units while PYTHIA is in event counts, a rescaling has to be applied before the two can be

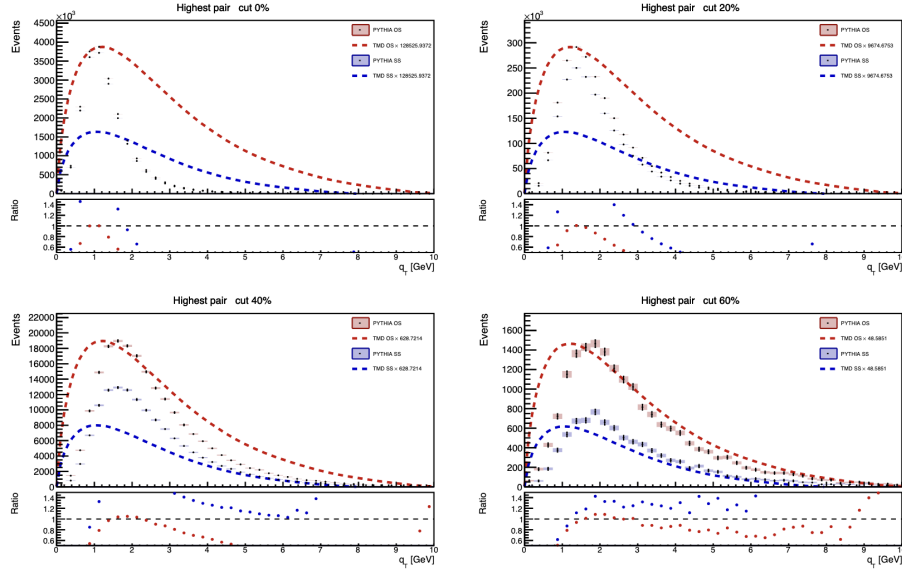


Figure 20: *Cut-dependent PYTHIA OS and SS q_T distributions with the independently generated TMD-theory curves overlaid after a common rescaling is applied to the OS and SS theory lines within each panel. The upper plots compare the shapes and relative OS/SS separation, while the lower plots show the bin-by-bin ratios of PYTHIA OS to theory OS and PYTHIA SS to theory SS respectively. The comparison is weakest for the uncut and 20% samples and becomes most reasonable for the 60% cut, especially above about 1 GeV.*

shown on the same plot. This rescaling is only used to compare the shapes of the distributions and the relative size of the OS and SS separation. It is not meant to claim a match in absolute normalization. The scale factor in each panel is found by taking the highest PYTHIA event count in that panel and dividing it by 30, since the standalone unscaled theory curve peaks at about 30 arbitrary units. That same scale factor is then applied to both the OS and SS theory curves in that panel. This is important, because using the same scale factor for both keeps the original theory separation between OS and SS unchanged. The scale factor used in each panel is given in the legend. For this overlay plot, the theory range is extended from 5 GeV to 10 GeV so that it can be compared more directly to the PYTHIA distributions.

The lower panels of Figure 20 are used to show more clearly how well the rescaled theory lines up with PYTHIA in each q_T bin. In these panels, the red points and blue points show the ratio of PYTHIA OS to TMD theory OS and PYTHIA SS to TMD theory SS respectively. A value close to 1 means that the PYTHIA count and the rescaled theory value are close in that bin. To keep the plots more readable, only ratio values up to about 1.4 are shown, so points that lie outside that range are not plotted. This is why some bins disappear in parts of the comparison, especially where the mismatch is larger.

The overall pattern of the comparison is clear. The agreement is poor for the uncut and 20% samples, improves at 40%, and is most reasonable for the 60% cut, especially for q_T above about 1 GeV. In the uncut and 20% panels, the theory does not line up well with PYTHIA either in the shape of the distributions or in the relative size of the OS and SS separation. This is visible both in the upper plots and in the

lower ratio panels, where many of the points sit far from 1 or fall outside the plotted range. At 40%, the comparison starts to improve, but the agreement is still not especially strong. By 60%, however, the comparison becomes much more reasonable. In that panel, both the OS and SS curves line up better in shape, and in the lower panel the PYTHIA-to-theory ratios stay within about 20% of much of the region above q_T around 1 GeV. This is not a perfect match, but it is clearly the best agreement of the four cuts. This pattern fits well with the picture from the previous section. At low cuts, the selected pions are less strongly tied to the original hard quark, since the sample still contains many softer pions coming from later shower and hadronization structure inside the jet. In that region, the flavour information from the initiating quark is more washed out, so the favoured versus unfavoured fragmentation difference is less clearly visible. At higher cuts, the selected pions are more leading in their jets and therefore remain more closely tied to the initiating quark. This is exactly the region where the fragmentation-function picture should be most visible. The fact that the theory comparison works best there strengthens the interpretation that the OS/SS difference seen in PYTHIA is really connected to the favoured versus unfavoured fragmentation structure.

It should also be noted that the best agreement appears for q_T values above about 1 GeV. However, this is still small compared to the hard scale of the process, since here Q is about 91 GeV. So even though the agreement is not concentrated only below 1 GeV, it still lies in the region where TMD sensitivity is expected to matter.

Overall, this section shows that the TMD theory by itself already predicts a clear OS/SS separation, and that this same pattern lines up best with PYTHIA only once the harder pion momentum-fraction cuts are applied. This strengthens the idea that the OS excess is not just a generic Monte Carlo feature, but is instead connected to the favoured versus unfavoured fragmentation picture, and that this becomes most visible when the selected pions are more leading in their jets.

4.3 Optimizing the comparison and confirming the conclusion

Having now identified the harder-cut region as the one where the OS/SS separation is clearest, the final step is to test how robust that conclusion is. The aim here is not to force an exact 1:1 PYTHIA–theory match, and it is also not to require a statistically “good” χ^2 value such as one close to 1. Instead, the purpose of this section is to check whether the comparison improves in the same region that the fragmentation picture would lead us to expect, and whether that improvement remains once reasonable changes are made to the setup. In that sense, this final section is mainly a consistency test. If the best-improving region is still the high- z , high-cut region, then that gives more confidence that the conclusion drawn in the previous

sections is not accidental.

To do this, two separate optimisation strategies are used. In both cases, the quantity being tracked is the combined χ^2 of the OS and SS comparisons. This is done so that the optimisation is not biased toward improving only one of the two distributions while making the other worse. A lower combined χ^2 means that the overall agreement has improved, while a higher value means that it has worsened. This combined χ^2 is used only as a relative marker of improvement or deterioration, not as a formal goodness-of-fit test. So although achieving a very low absolute χ^2 would of course not be a bad thing, that is not the main goal here. The main question is simply whether the comparison improves in the physically expected direction.

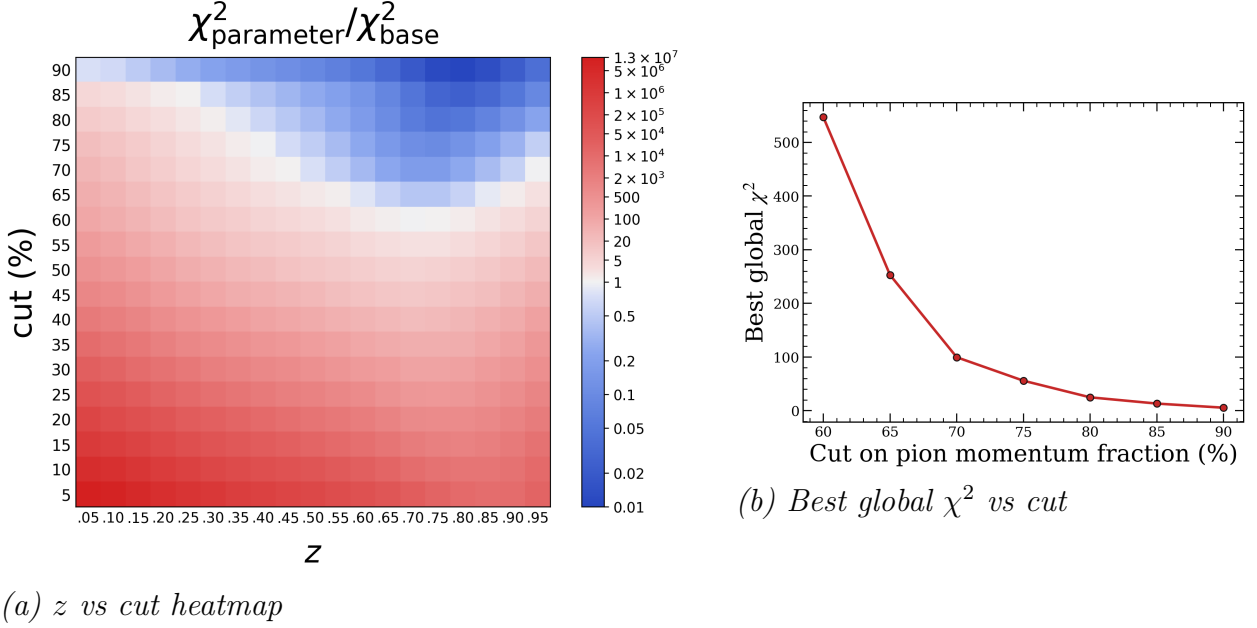


Figure 21: Optimisation scan in z and pion momentum-fraction cut. The left panel shows the ratio $\chi^2_{\text{parameter}}/\chi^2_{\text{base}}$ across the scanned (z, cut) region, where blue indicates an improvement relative to the baseline 60%, $z = 0.70$ comparison and red indicates a worse comparison. The right panel shows, for each cut, the lowest combined χ^2 obtained after scanning over z .

The first strategy is to test more directly whether the earlier interpretation of high z and hard pion momentum-fraction cuts is really correct. This is done by scanning z from 0.05 to 0.95 in steps of 0.05, and comparing each of these theory settings against cuts from uncut up to 95%. The result is shown in Figure 21. In the left panel, the horizontal axis is z and the vertical axis is the pion momentum-fraction cut. Each square shows the ratio $\chi^2_{\text{parameter}}/\chi^2_{\text{base}}$, where the base value is the original 60%, $z = 0.70$ comparison. A blue square therefore means the new choice gives a lower combined χ^2 than the base result, while a red square means the comparison has become worse. White indicates values close to the base result.

The main thing to notice in this heatmap is that the more favourable region does indeed move toward harder cuts and relatively high z , which is exactly what the earlier fragmentation-based interpretation would suggest. The smaller- z , smaller-cut region is mostly red, meaning that these combinations generally

worsen the comparison. By contrast, once the scan moves toward higher z and harder cuts, much more blue begins to appear. This supports the idea that the part of the PYTHIA sample which lines up best with the theory is the one where the selected pions are more leading and therefore more closely tied to the original quark fragmentation picture.

There is, however, one important caveat. The very hardest-cut corner, especially above about 85%, appears artificially too favourable. Although these squares are among the darkest blue in the heatmap, this is mainly due to the fact that the total event count has become very small and the uncertainties have grown unnaturally large. In that situation the combined χ^2 can look much better than it really is, simply because the comparison has become weakly constrained. For that reason, the extreme-cut corner is not taken as physically meaningful evidence by itself. The more reliable takeaway is that, within the statistically useful region, the improvement clearly moves toward harder cuts and relatively high z .

This same scan is summarized in the right panel of Figure 21. For each cut, the z value that minimizes the combined χ^2 is selected, and the resulting best global χ^2 is then plotted as a function of the cut. This makes the overall trend easier to see than in the full heatmap alone. One important thing this graph shows is that although the 60%, $z = 0.70$ case was the first one to give a convincing theory comparison in the earlier plots, it is not the absolute optimum of the full scan. The best global χ^2 improves further at 65% and 70%, and continues to fall at even harder cuts. Again, however, the strongest improvements at the very highest cuts are not taken too literally because they are heavily influenced by the loss of statistics. What matters more is that the scan confirms the original physical picture: the comparison improves once the analysis is pushed toward high z and harder momentum-fraction cuts.

It is also worth noting that the absolute χ^2 values remain very large, with the baseline result around 550. This is exactly why the combined χ^2 is not being used here as a formal statement that the theory gives a statistically good fit. The useful information is the relative improvement. For example, the reduction by about a factor of five by the time the scan reaches around 70% is meaningful as evidence that the comparison is improving in the expected direction, even though the absolute χ^2 is still far too large to claim a full fit. In that sense, the 60%, $z = 0.70$ point should not be viewed as the unique optimum, but rather as the first clear instance where the PYTHIA sample begins to show the same favoured versus unfavoured fragmentation picture as the theory.

This is shown more directly in Figure 22. For the 65% and 70% cuts, the best agreement is found at $z = 0.70$, while for the 75% and 80% cuts it shifts to $z = 0.75$. This is a useful result, since it shows that the region giving the strongest comparison is not random, but sits in a consistently high- z range. This fits naturally with the rest of the analysis. The harder cuts are already selecting more genuinely leading

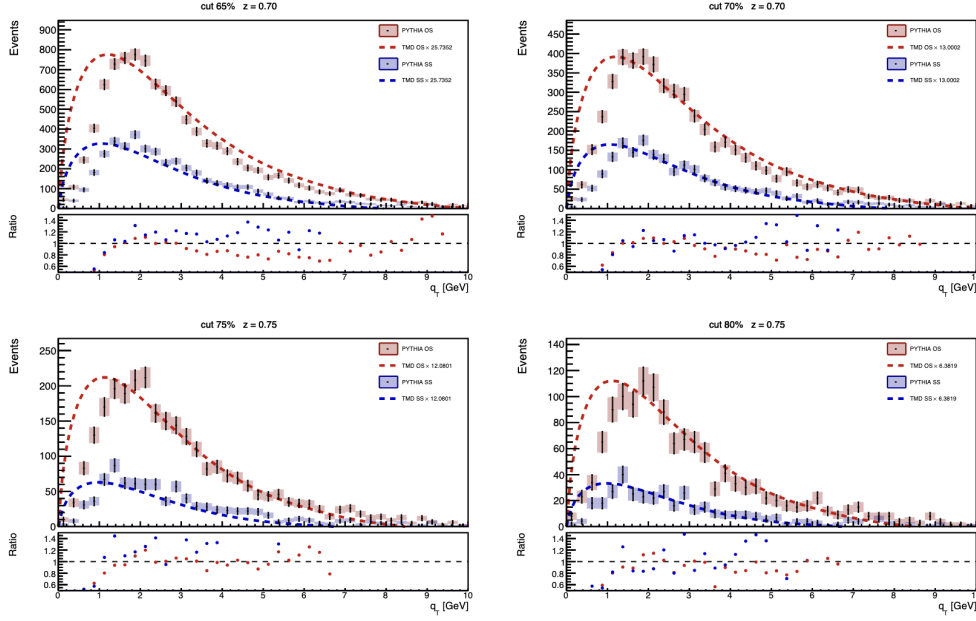


Figure 22: *Overlay plots for the best- z points at the hardest pion momentum-fraction cuts in the scan. For each cut shown, the value of z is chosen to minimize the combined OS+SS χ^2 . These plots show more directly how the apparent improvement evolves as the cut is pushed harder, and also why the very highest-cut region must be treated carefully once the event count becomes too small.*

pion pairs, so it is reasonable that the best theory comparison should also move toward a more leading fragmentation regime. The important point is not that this gives a final extracted z , but that it identifies the region where the Monte Carlo and theory are most closely aligned within the present setup.

The second strategy is to test whether changing the PYTHIA parameters can improve the correlation with the theory. As mentioned earlier in the Monte Carlo section, PYTHIA is a phenomenological model tuned to real data rather than something derived directly from the same TMD framework as the theory curves. Because of this, it is reasonable to ask whether modest parameter changes might make the generated event sample line up more closely with the TMD prediction. Two parameters were chosen for this scan: the timelike-shower α_s , which affects the shower stage, and `StringPT:sigma`, which affects the transverse-momentum scale in the hadronization stage. These two parameters were chosen because they change different parts of the event development and therefore give a useful first test of whether the mismatch is sensitive to the Monte Carlo setup itself.

The result of this scan is shown in Figure 23. The horizontal axis is `StringPT:sigma` and the vertical axis is the timelike-shower α_s value. Each square again shows the ratio of the new combined χ^2 to the default 60%, $z = 0.70$ baseline. Blue means the comparison has improved and red means it has worsened. Since changing the parameters alters the full event generation, each square in this heatmap requires the whole PYTHIA simulation to be rerun. Because of this much larger computational cost, a reduced event sample of 20 million events was used for this scan rather than the full 100 million used earlier.

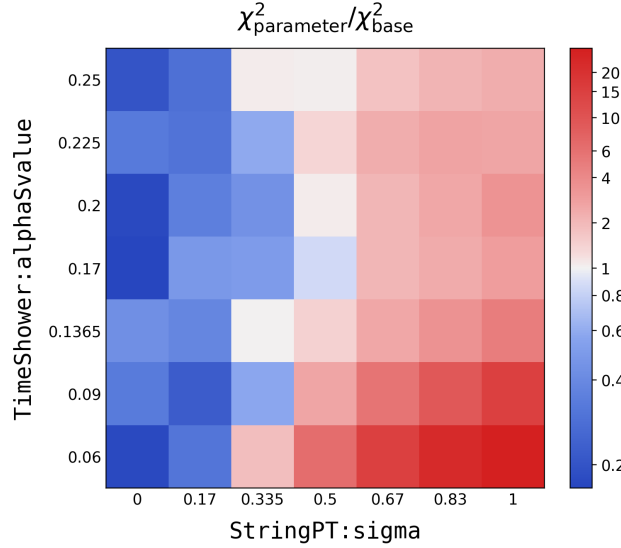


Figure 23: Heatmap showing how the comparison quality changes when *StringPT:sigma* and the timelike-shower α_s are varied in PYTHIA. Each square shows the ratio of the new combined OS+SS χ^2 to the default 60%, $z = 0.70$ baseline. Blue indicates an improvement, while red indicates a worse comparison.

The main thing to notice in Figure 23 is that the timelike-shower α_s does not produce a very strong overall pattern, while *StringPT:sigma* has a much larger effect on the comparison. Much of the more favourable region appears toward the lower values of *StringPT:sigma*, while larger values tend to worsen the fit. This suggests that the hadronization transverse-momentum scale is the more important of the two scanned parameters for this particular comparison. However, the heatmap alone does not tell the full story, since an apparent improvement in the combined χ^2 may still come from an undesirable change in the event sample. For this reason, the selected overlay plots are shown in Figure 24. These make it clearer what is actually happening behind the heatmap. The main pattern is that many of the apparent improvements are driven by large decreases in the total event count, rather than by a uniformly better physical agreement across the whole q_T range. This effect is not simply due to the reduced 20 million event sample used in the scan, but comes from the parameter changes themselves, which alter the generated event structure. In that sense, not every lower combined χ^2 should be interpreted as equally meaningful.

At the same time, Figure 24 also shows that some of the parameter changes do genuinely improve the agreement in the very low- q_T region, especially below about 1 GeV. This is encouraging, since it shows that PYTHIA parameter changes can move the comparison in the right direction. However, this gain is not uniform. In several of these improved parameter points, the better agreement below 1 GeV is accompanied by a weaker comparison above 1 GeV, which is exactly the region where the earlier TMD comparison had looked most promising. Because of this trade-off, the result of the parameter scan is best described as suggestive rather than conclusive. It shows that parameter variation has the potential to strengthen the PYTHIA–theory comparison, but it does not yet give a clearly improved global picture in the physically

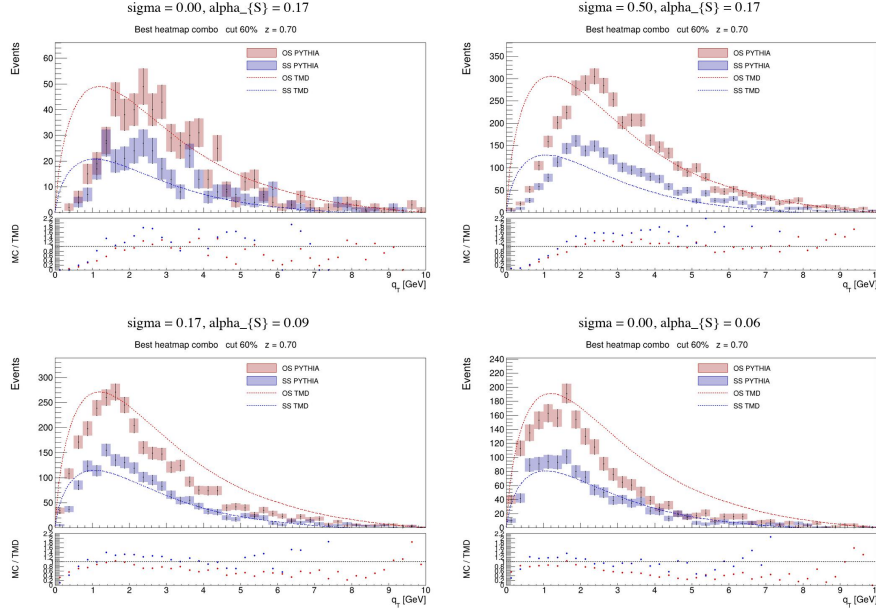


Figure 24: Representative overlay plots for four improved points from the *PYTHIA* parameter scan. The titles give the chosen values of *StringPT:sigma* and the timelike-shower α_s . These plots show that some of the apparent improvement in the heatmap corresponds to a real change in the *PYTHIA*–theory comparison, but also that part of the gain can come from a large reduction in the total event count rather than a uniformly better match across the full q_T range.

most useful region.

Taken together, these two optimisation strategies strengthen the overall interpretation of the results, even though neither is being used to claim a final fit. The z and cut scan supports the original idea that the theory comparison improves in the high- z , hard-cut region, which is exactly where the selected pion pairs should be most leading and most closely tied to the initiating quark. The parameter scan then shows that the comparison is also sensitive to the details of the Monte Carlo modelling, especially in the hadronization stage, although these changes do not yet produce a robust overall improvement.

5 Conclusions

This thesis set out to test whether charged-pion pairs selected from opposite jets in e^+e^- collisions can define a useful TMD-sensitive observable. More specifically, the goal was to determine whether the transverse momentum of the selected pion pair, together with the difference between opposite-sign and same-sign yields, could reveal fragmentation structure in a way that is both physically meaningful and comparable to TMD-based theory. The analysis was restricted to simulation and theory, with the longer-term motivation of identifying a region that could later be used in a data-driven study.

The first main result is that the observable only becomes physically useful once the event axis and event selection are chosen carefully. Defining the pion-pair transverse momentum with respect to the thrust

axis, and imposing cumulative cuts on the pion momentum fraction, isolates a much cleaner leading-pion, back-to-back dijet configuration than less selective choices. In this setup, a clear OS/SS separation appears in the PYTHIA results, and this separation strengthens systematically as the pion momentum-fraction cut is increased. At the same time, the selected distributions broaden toward somewhat larger q_T , showing that the harder cuts are not just removing events, but are isolating the part of the sample that is more closely tied to the original fragmentation process.

The second main result is that this OS/SS trend is qualitatively supported by the independent TMD-based theory calculation. The standalone theory curves already show a clear OS/SS separation at high z , and once they are overlaid on top of the PYTHIA results, the most reasonable agreement is found in the harder-cut region, especially for the 60% sample and above about 1 GeV. This does not amount to a full fit, and the agreement is not uniform across the whole q_T range, but it is strong enough to support the underlying physical picture. In particular, it shows that the OS excess seen in the Monte Carlo is not arbitrary, but is consistent with the favoured versus unfavoured fragmentation structure built into the TMD theory.

The optimisation studies then strengthen this interpretation rather than replace it. The (z, cut) scan shows that the comparison improves in the high- z , high-cut region, which is exactly where the selected pion pairs should be most leading and most closely tied to the initiating quark. The PYTHIA parameter scan shows that the observable is also sensitive to the details of the Monte Carlo modelling, especially in the hadronization stage, although these changes do not yet produce a robust global improvement. These studies therefore do not change the main conclusion of the thesis, but they do show that the preferred region is not accidental and that the comparison can be improved in a controlled way.

The main conclusion is therefore not that a full extraction of pion TMD fragmentation functions has already been achieved. Rather, this thesis has identified and validated the region in which such an extraction would be most promising. Within the present setup, that region is the one where the selected pion pair is more leading, the OS/SS separation is clearest, and the comparison to the independent TMD theory is most reasonable. Further progress would come from broader theory scans, improved optimisation of the Monte Carlo description, and eventually comparison to archival or future e^+e^- data.

For a more in-depth view of the progress of the project as a whole, including the code and weekly presentation slides developed throughout it, the reader is referred to the project repository at <https://github.com/Shane-Sweetman/TMD-Analysis/tree/main>.

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